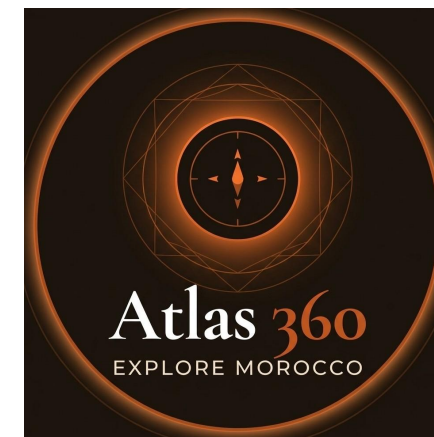


Atlas 360 — A Digital Marketplace for Curated Travel and Local Guides in Morocco



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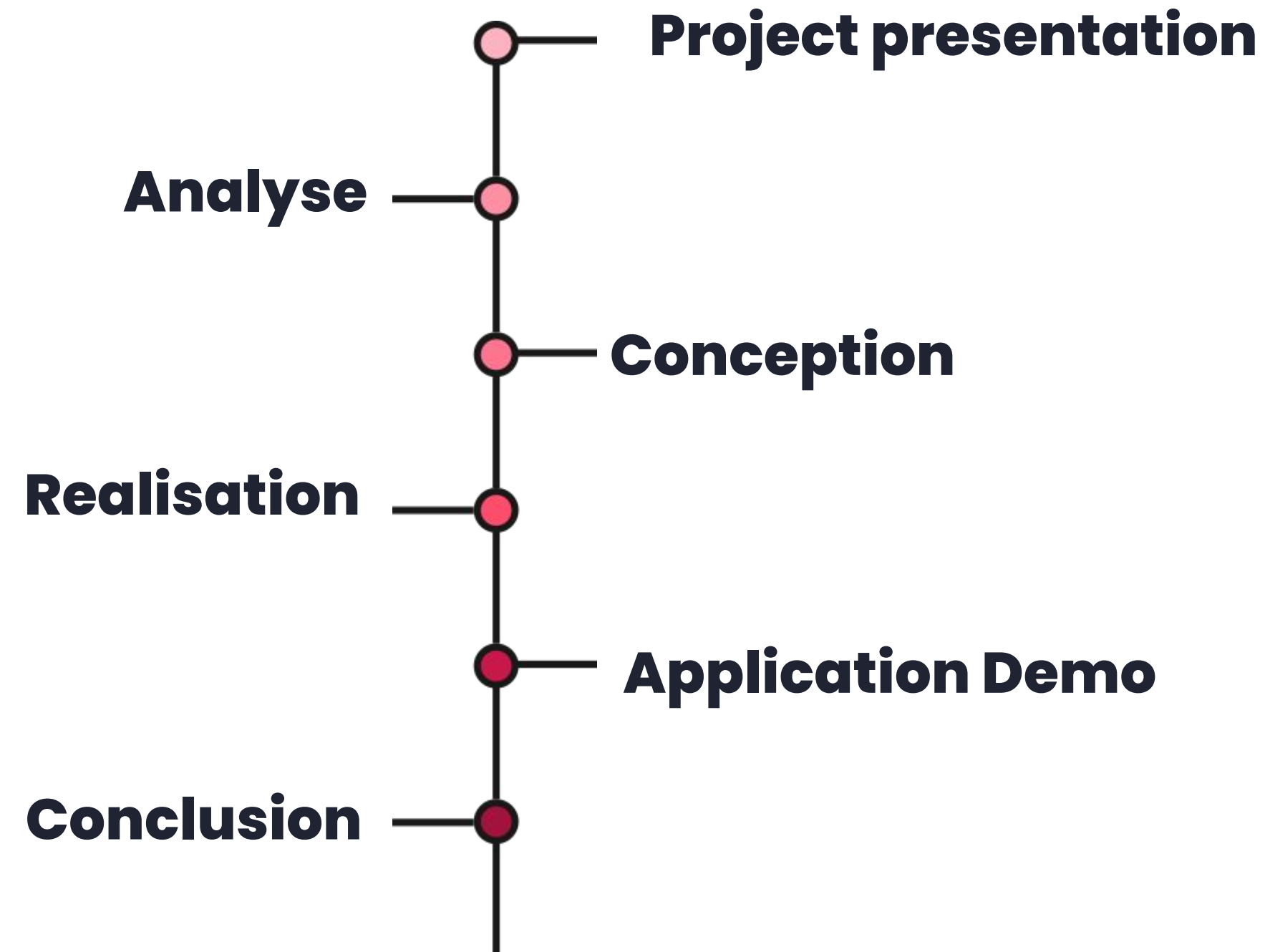


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A.U. 2025-2026

ENCADRÉ PAR :
Azeddine
Elhassouny

Plan de **Présentation**





Introduction



01

Project Presentation

Context & *Problematic*



Traveller Friction

- **Fragmented Booking:** No central directory for remote excursions, leading to disconnected routes.
- **Unverified Safety:** Lack of verification for high-altitude guides exposes travellers to extreme trail risks.
- **Static & Blind Trails:** Standard map providers fail on Saharan dunes and unpaved mountain pathways.



Guide Barriers

- **Offline Exclusion:** Skilled Berber and Sahrawi guides remain completely invisible in modern digital ecosystems.
- **Administrative Vacuum:** Lack of digital schedules leads to double-bookings and scheduling friction.
- **Payment Insecurity:** Complete reliance on cash-on-arrival triggers high last-minute cancellation risks.



API Cost Overhead

- **Naive AI Traps:** Raw LLM conversational builders make heavy, unfiltered calls back to heavy spatial servers.
- **Runaway Token Fees:** Frequent map rendering prompts can balloon OpenAI / Google Cloud API budgets exponentially.
- **Economic Instability:** Uncontrolled queries degrade platform sustainability in thin-margin local travel tiers.

OBJECTIVES

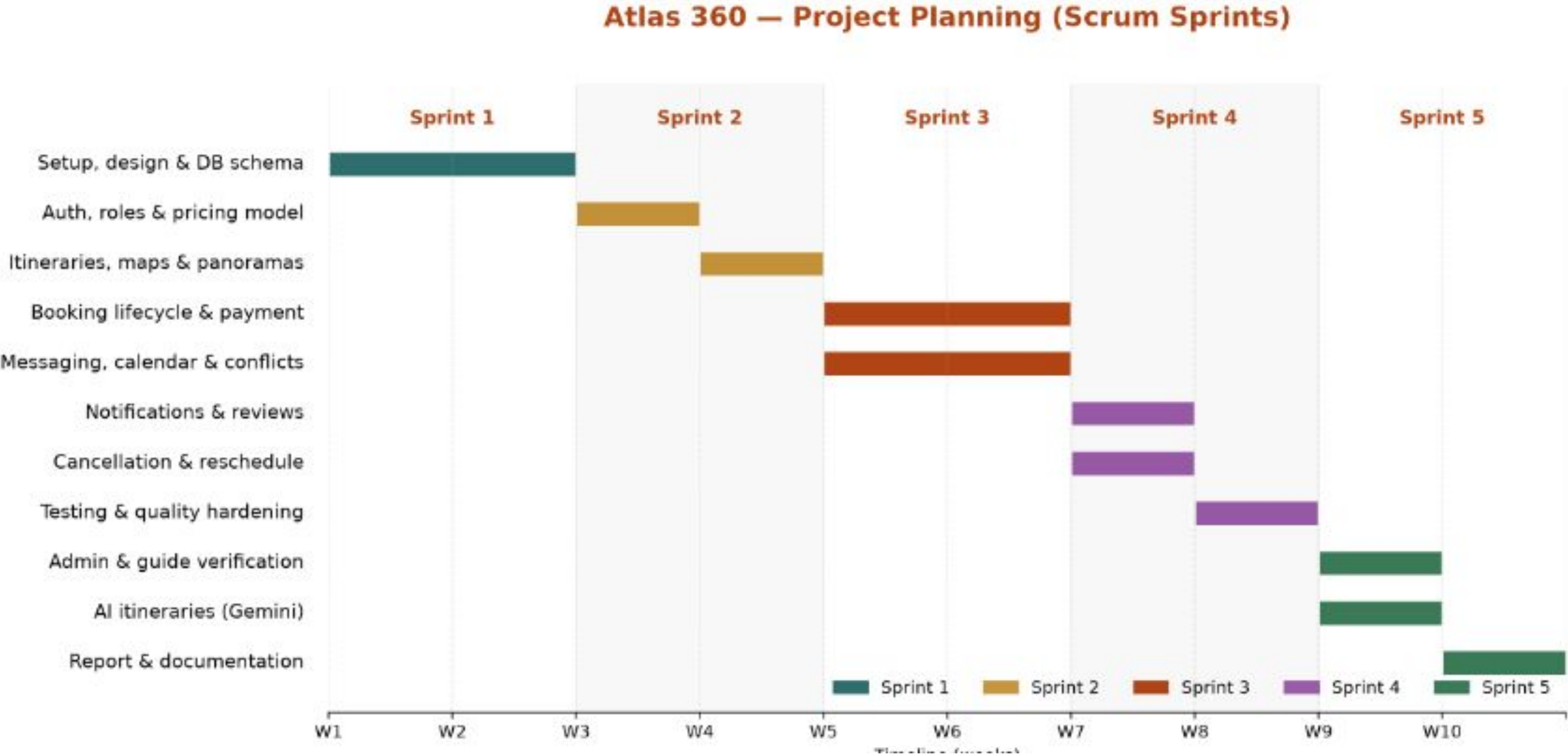


- offers curated itineraries enriched with interactive maps and 360° panoramas;
- proposes a clear subscription model that unlocks premium content;
- provides a marketplace where verified local guides can be discovered and booked;
- secures every booking through a complete lifecycle and an escrow logic;
- gives administrators the tools to verify guides, release payouts and moderate content.



Methodology and Project Organisation





02 Analysis

Functional Requirements

Functional Requirements

- **Visitor:** Can explore itineraries and guide profiles, access limited content, and create or access an account.
- **Traveller:** Can subscribe to premium plans, access all travel content, book and pay for guides, communicate with them, manage bookings, and leave reviews.
- **Guide:** Can manage their profile and availability, handle booking requests, communicate with travellers, receive payouts, and track performance metrics.
- **Administrator:** Oversees platform operations by verifying guides, managing payments and reviews, handling support requests, and administering user roles.

Non-Functional Requirements

Security: Authentication, role-based access control, row-level database security, and protection of sensitive operations.

Data Integrity: Prevent double-booking of guides, even during concurrent requests.

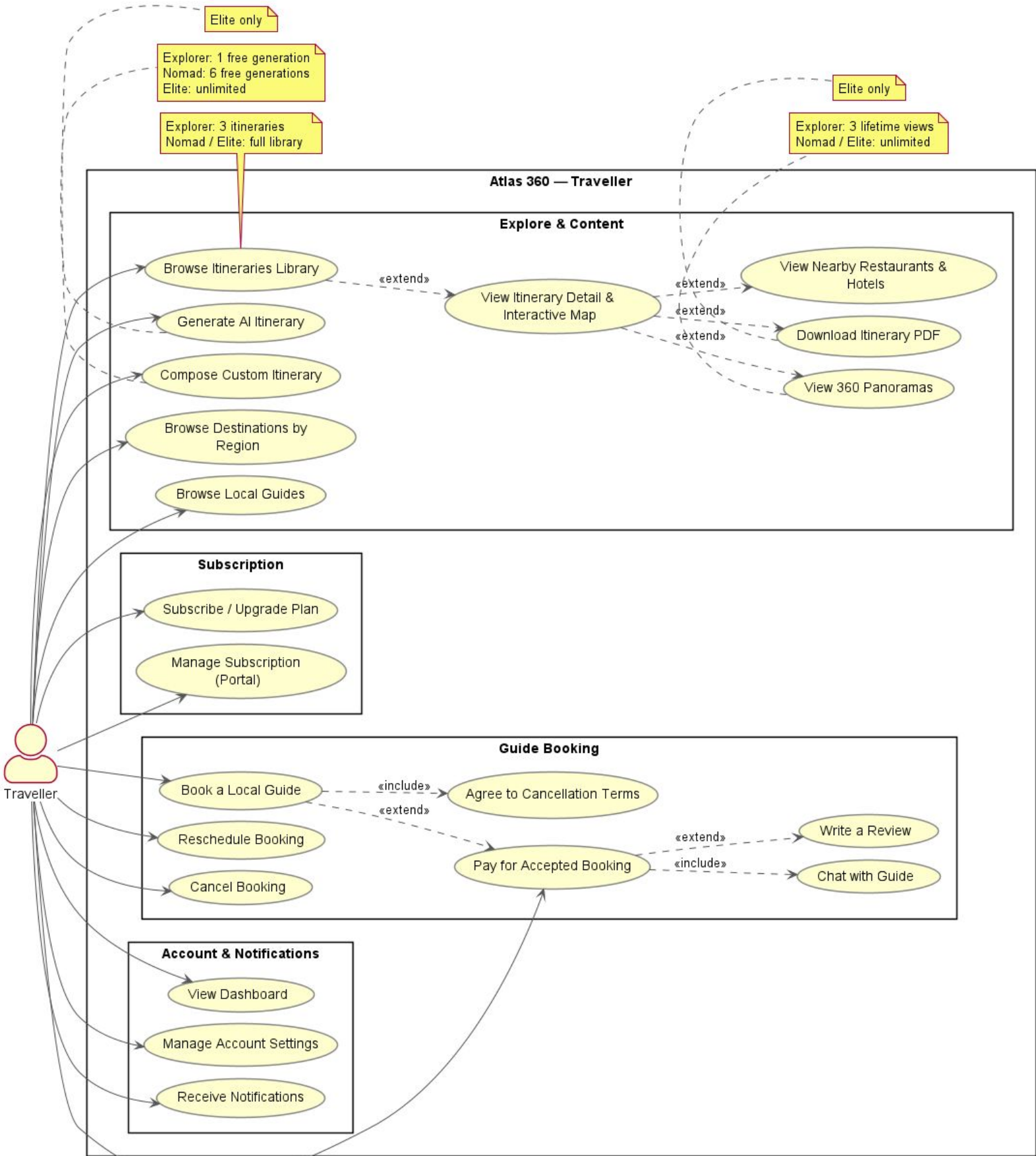
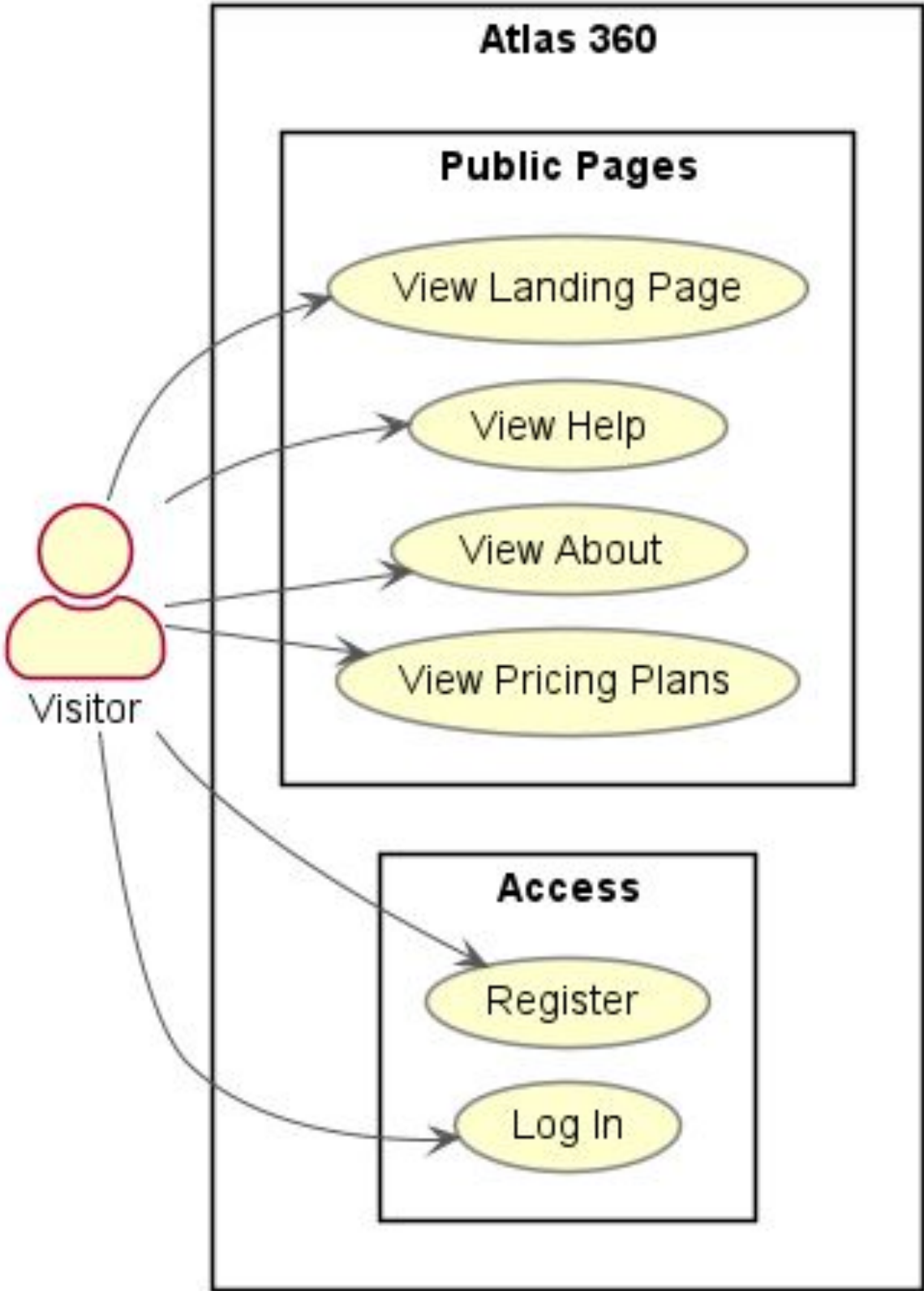
Transaction Reliability: Hold payments securely until service completion (escrow system).

Usability: Clean, responsive, and consistent interface with light and dark modes.

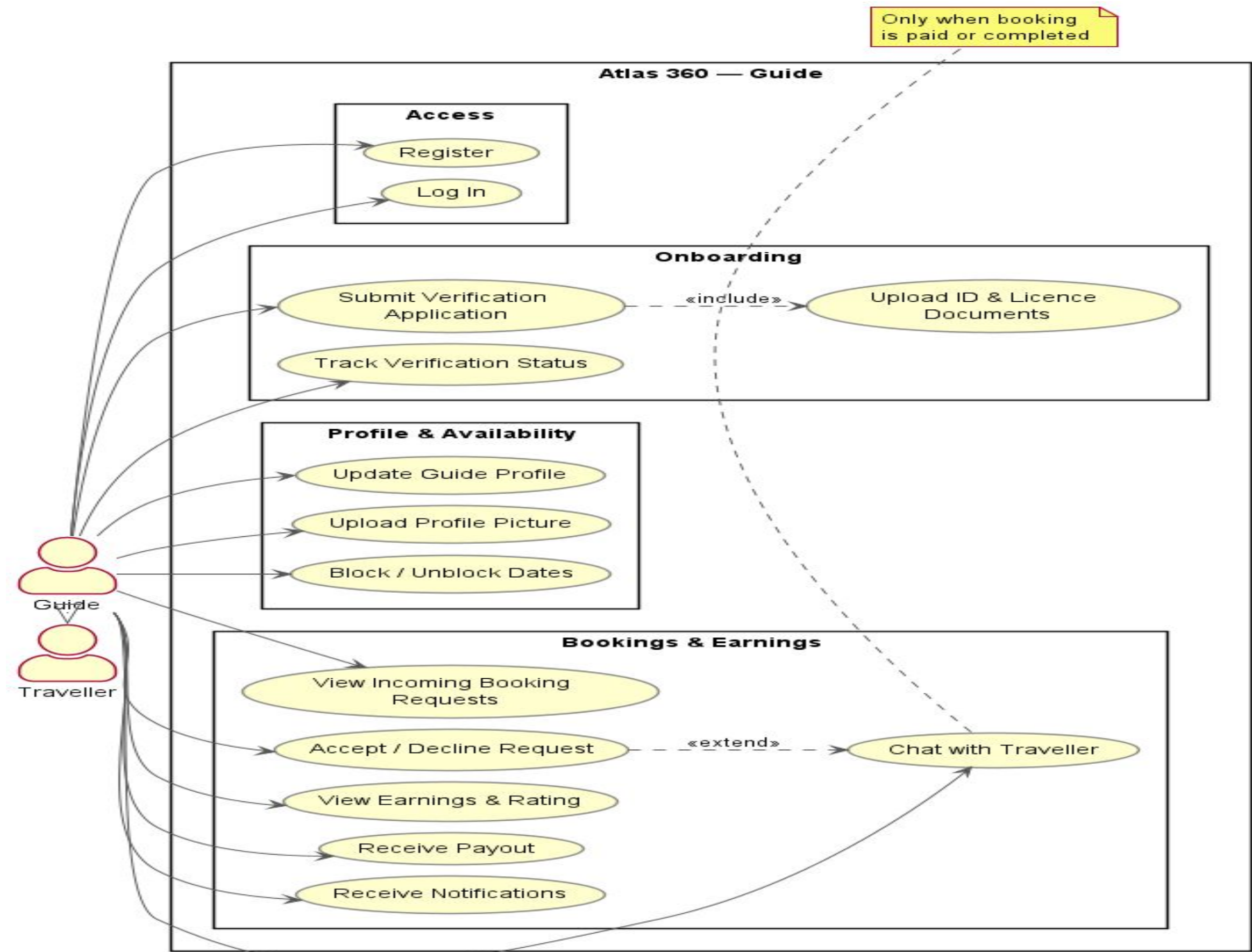
Maintainability: Modular architecture and automated testing.

Performance: Fast response times and server-side rendering support.

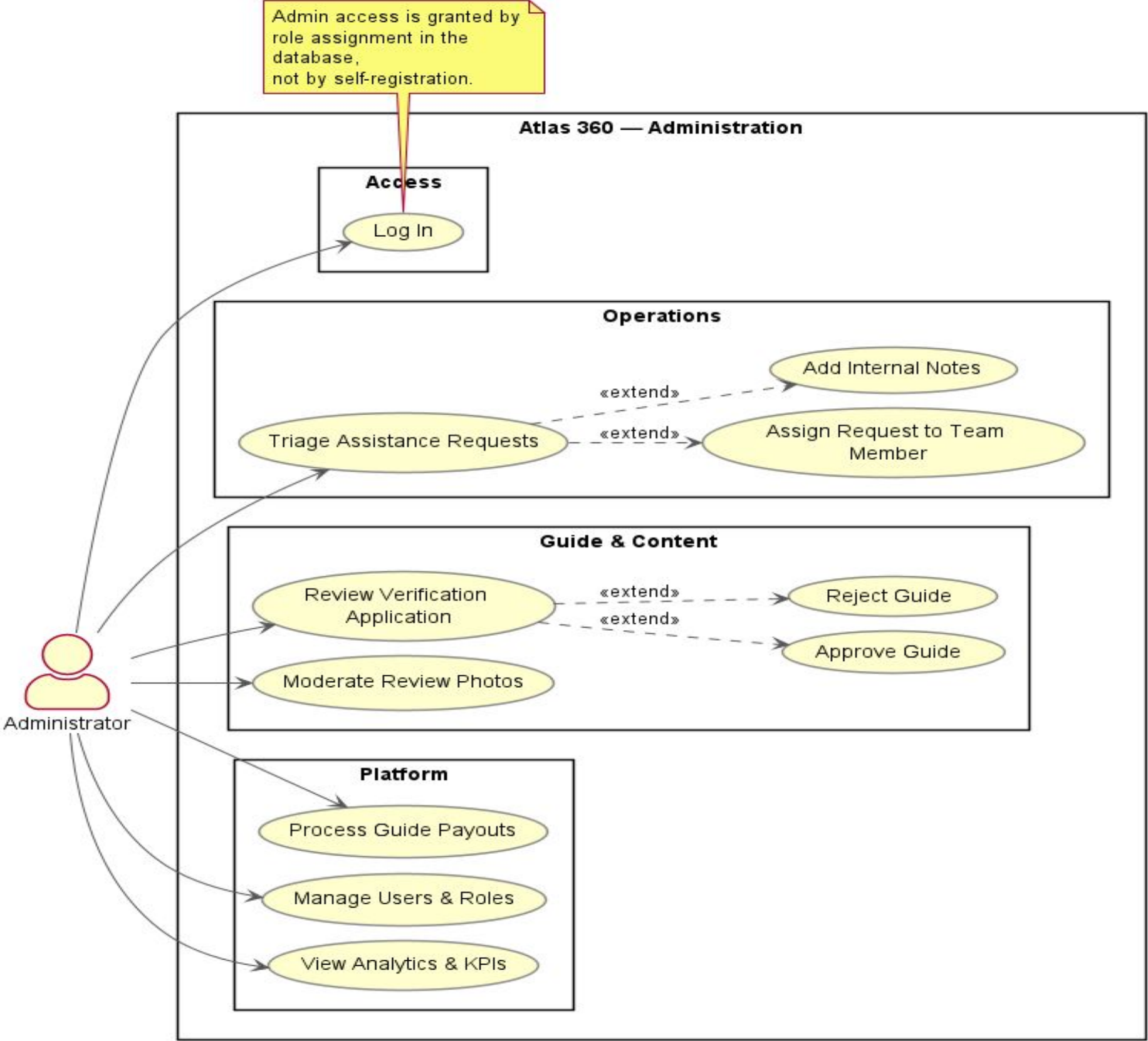
USE CASE DIAGRAMS



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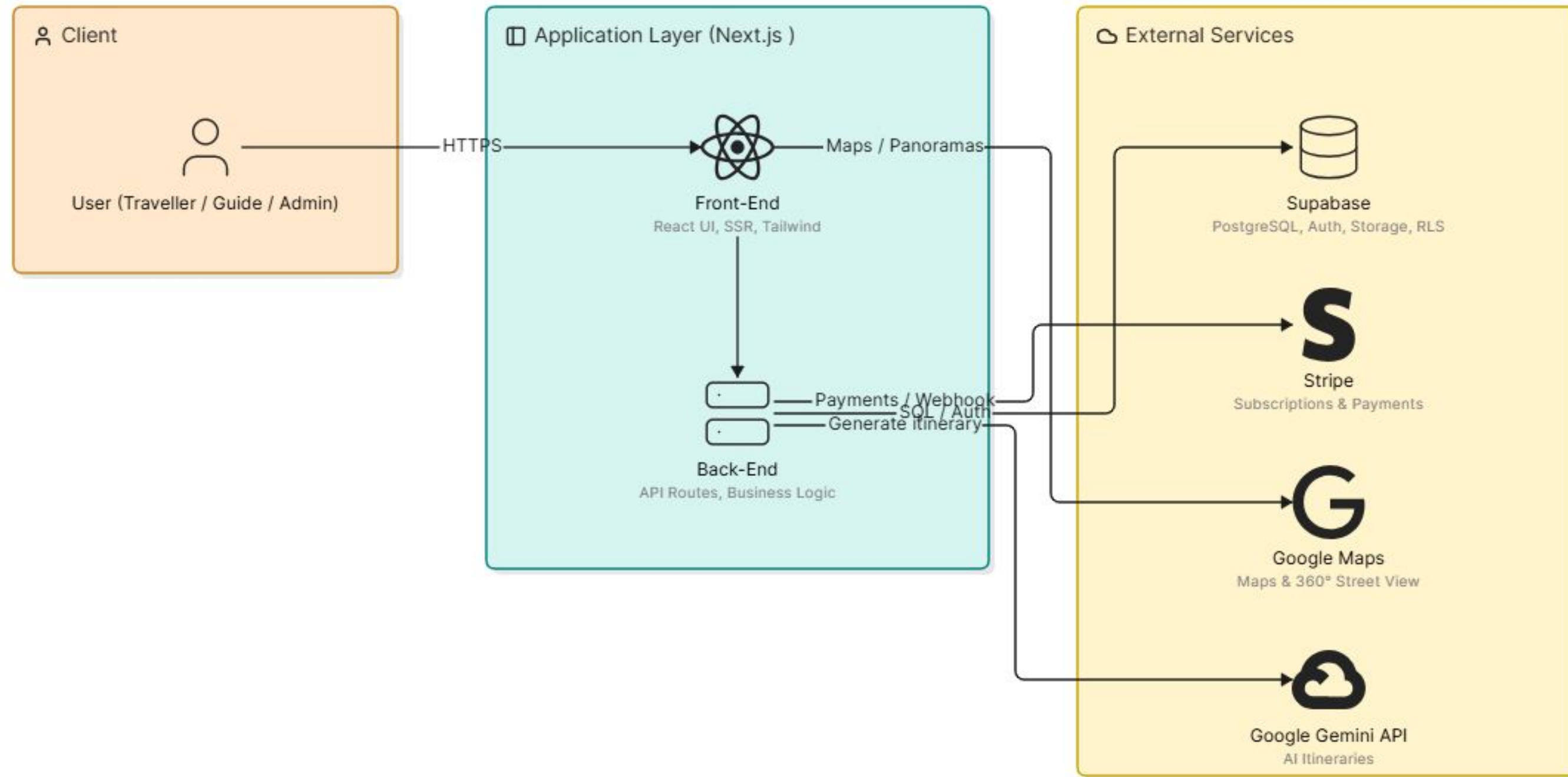


USE CASE DIAGRAMS

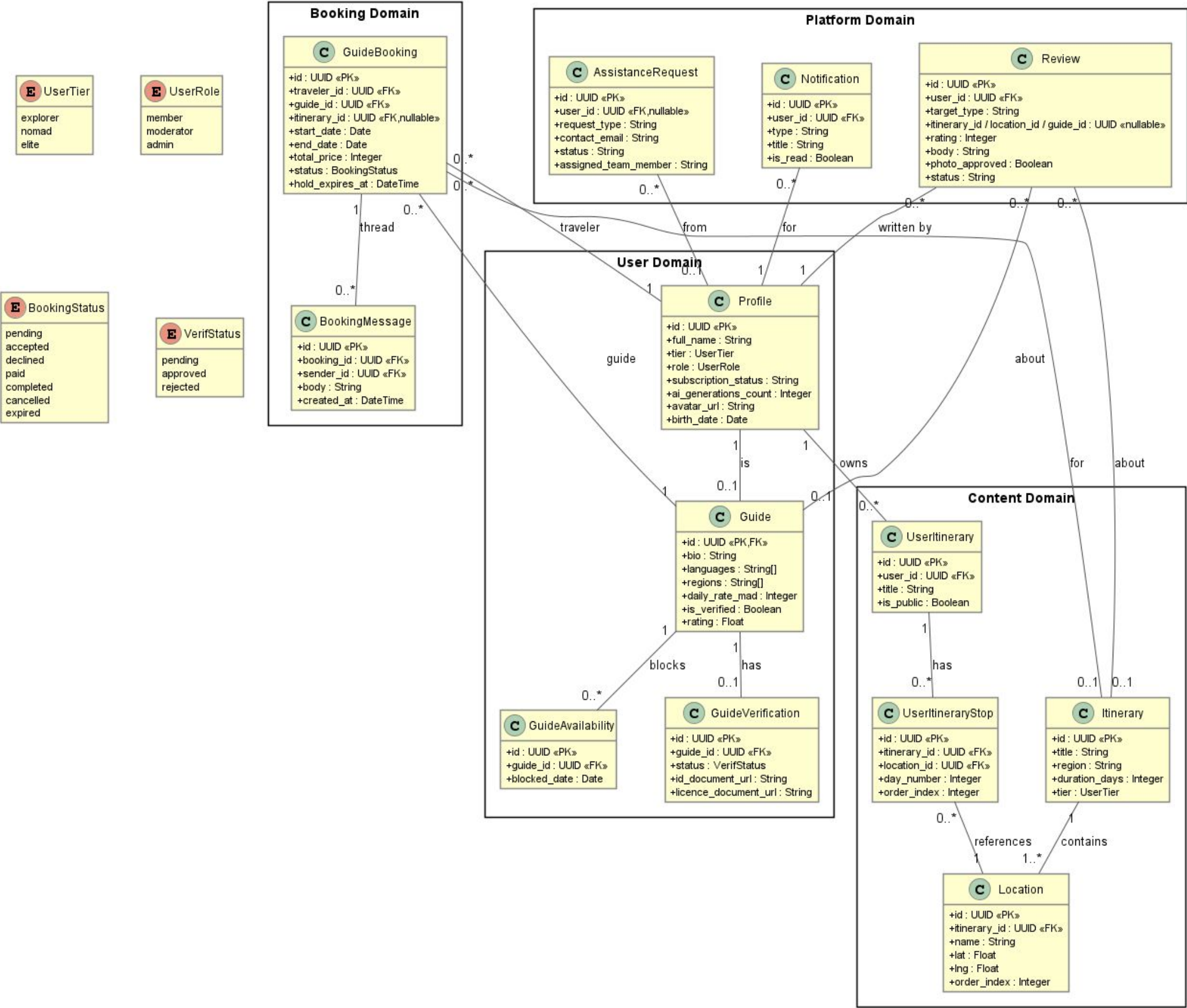


03 **Conception**

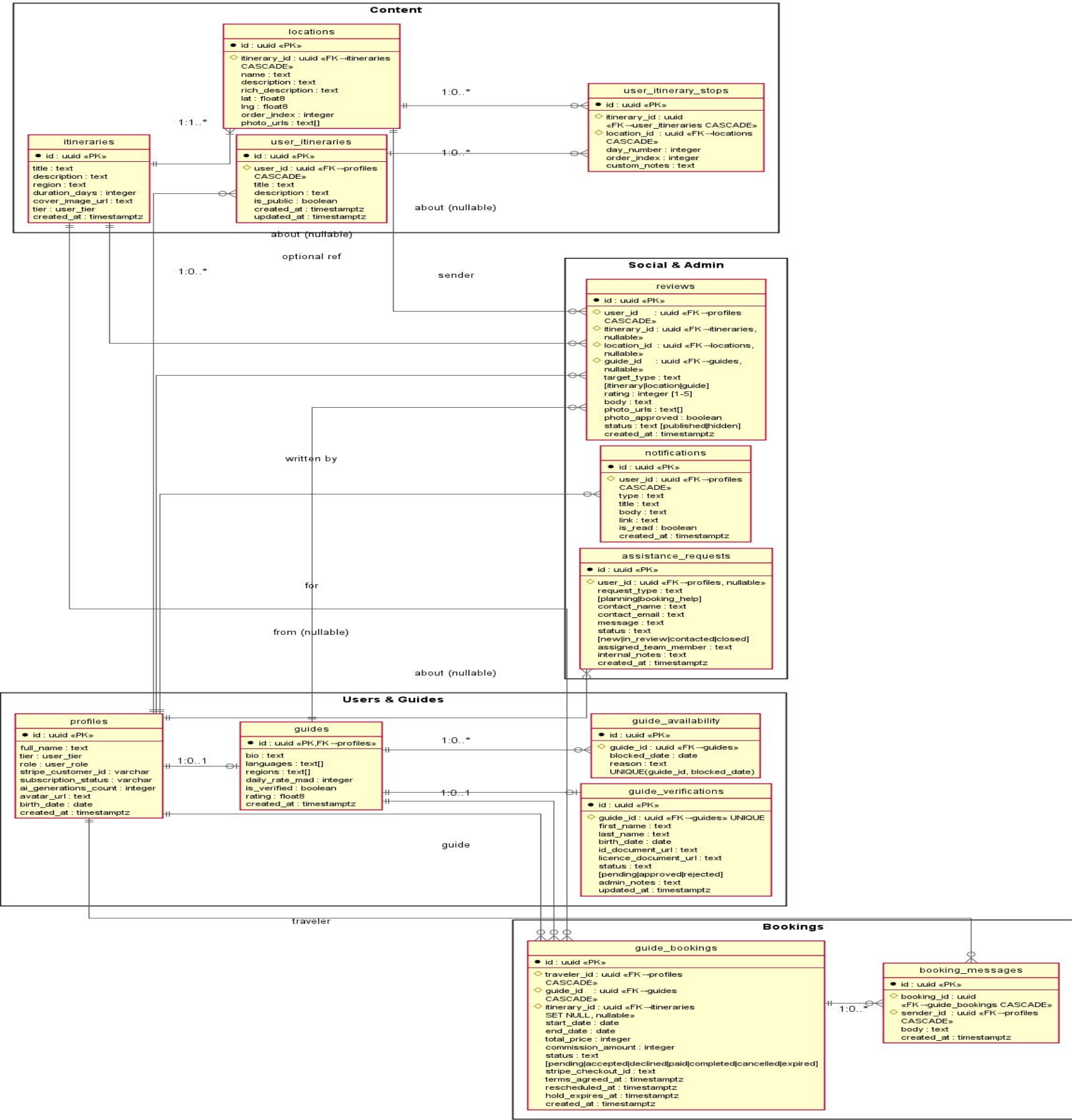
Conceptual Design & System Architecture



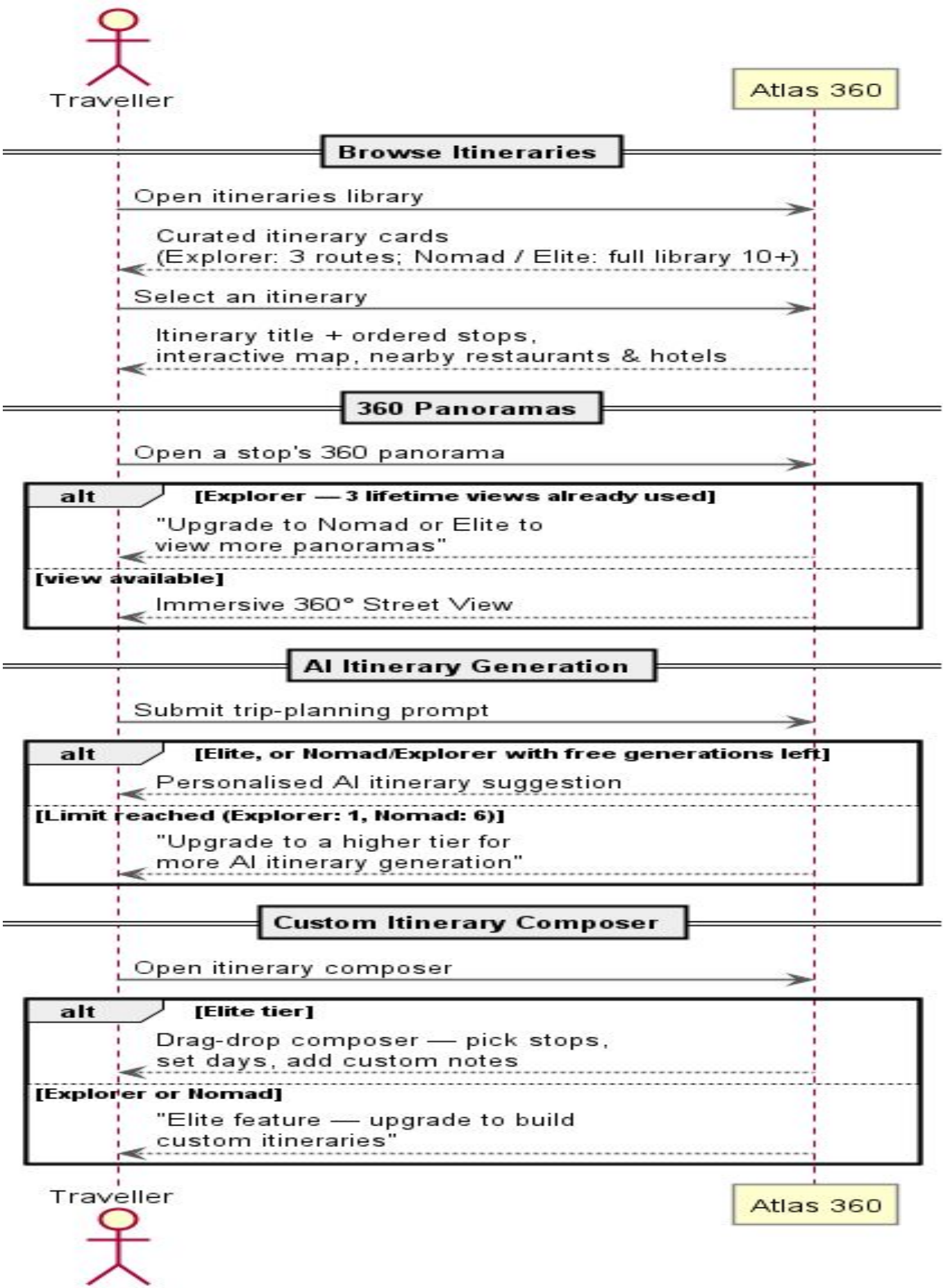
Classe Diagram



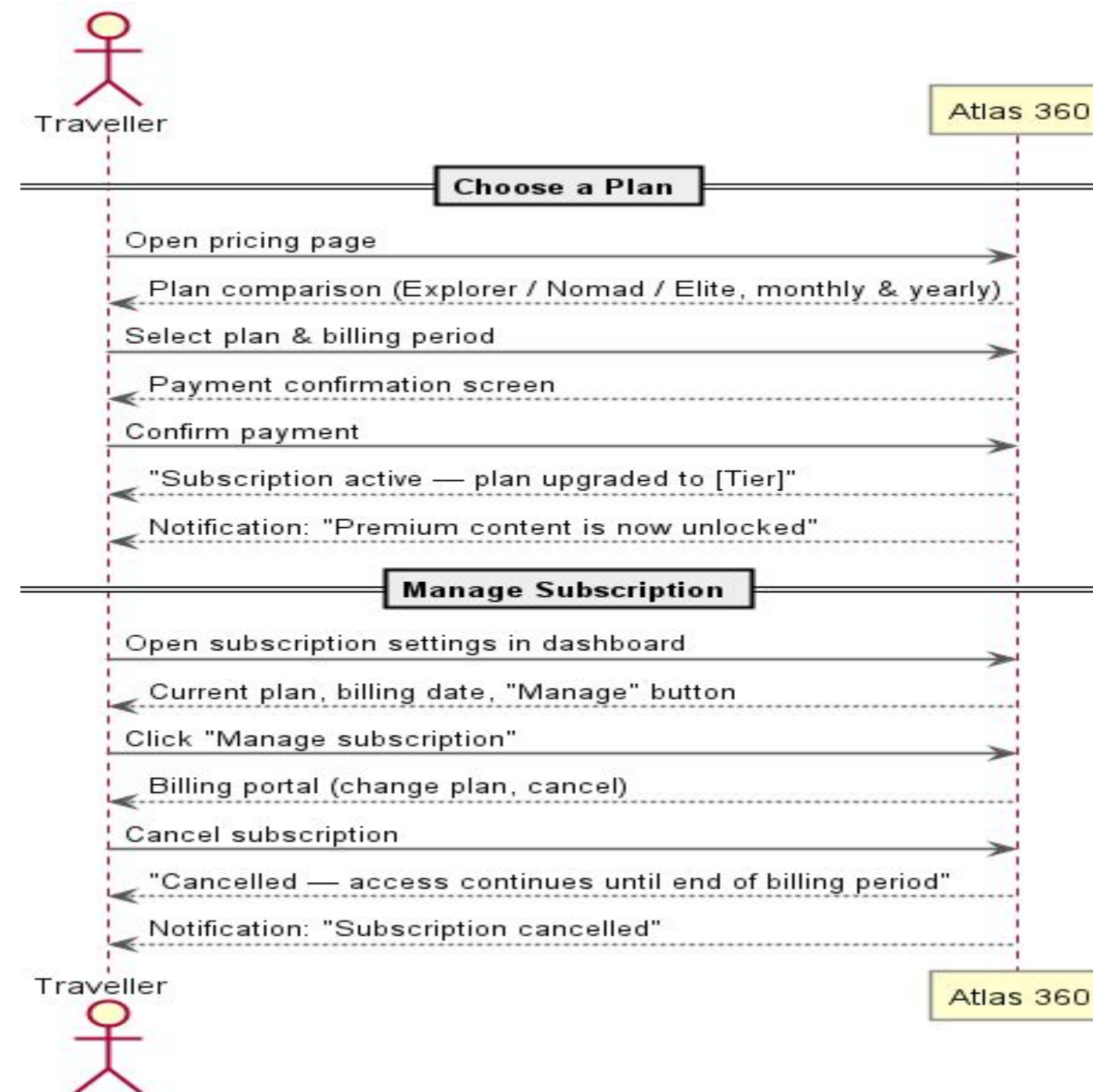
Data Model



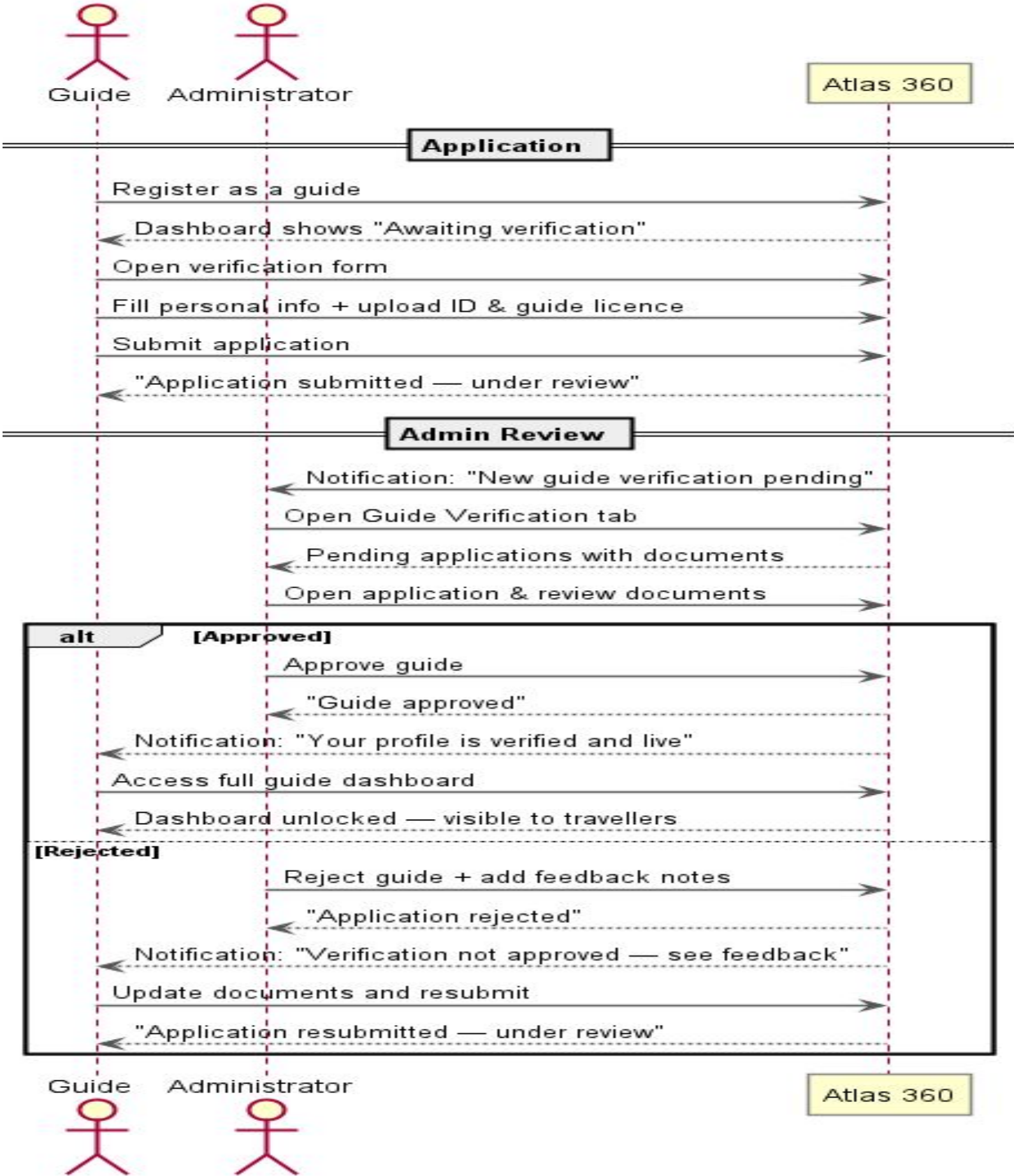
Sequence Diagram
Explore



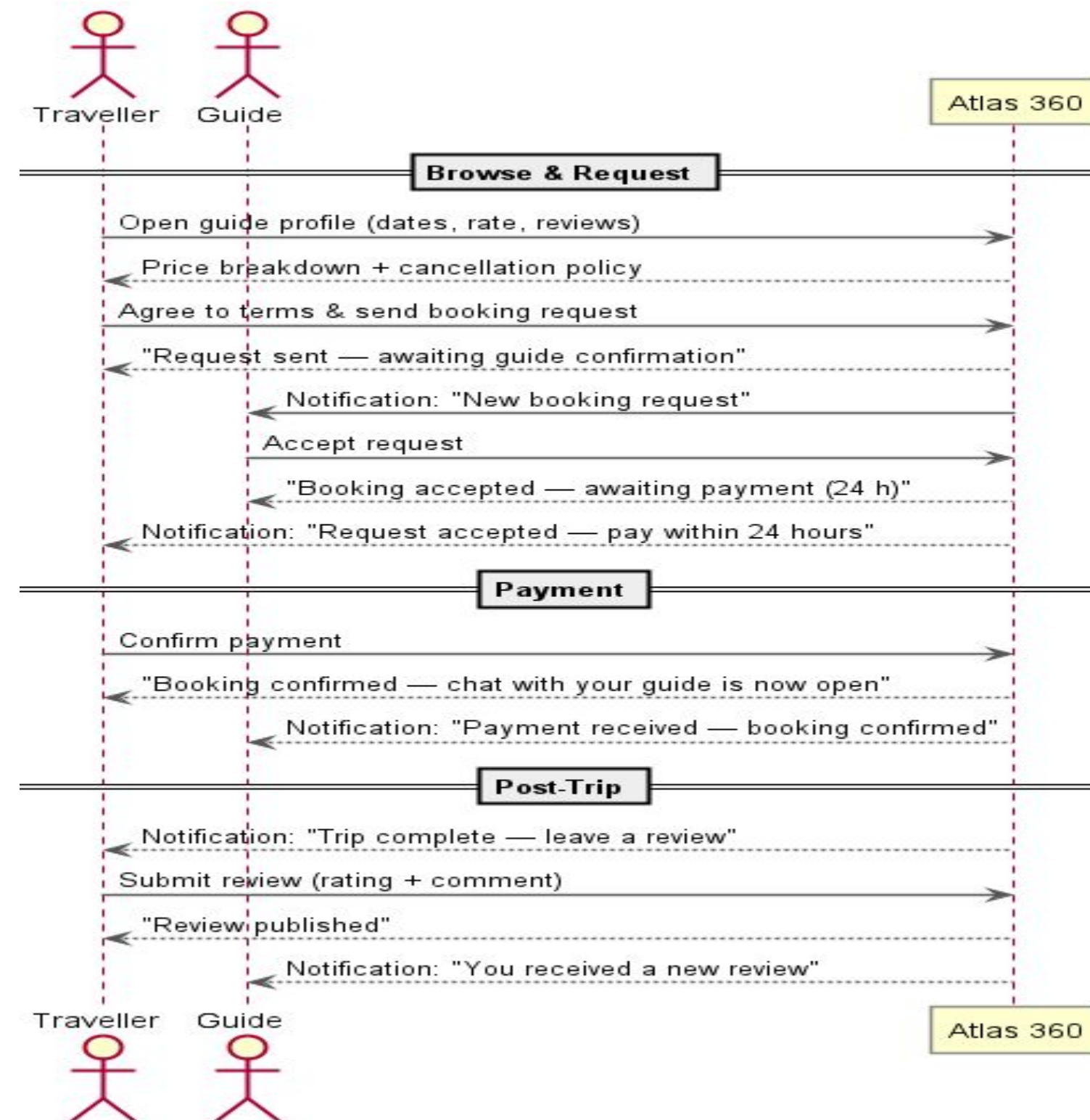
Sequence Diagram Subscription



Sequence Diagram
Guide Verification



Sequence Diagram Guide Booking



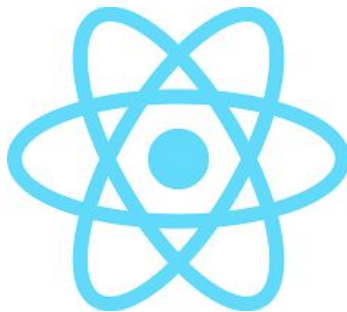
04

Implementation

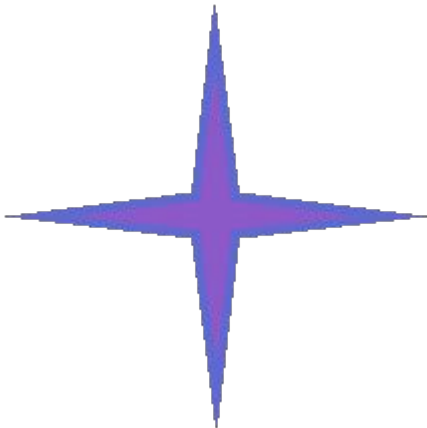
**Development
Environment and Tools**

App Demo

Languages and Front-end Framework



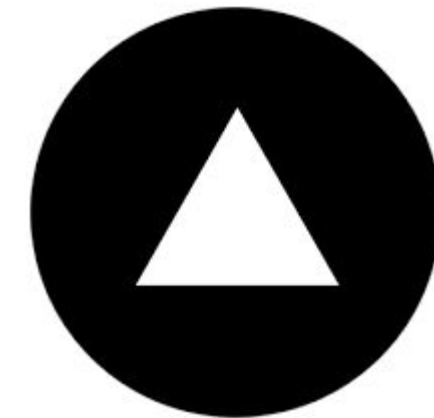
Back-end, Database and Services



**Development
Environment and Tools**

App Demo

Hosting, Runtime and Quality



Development
Environment and Tools

App Demo

Financial Viability

Testing and Quality
assurance



Development
Environment and Tools

Démonstration du Plateforme de
Monitoring

Financial Viability

Testing and Quality
assurance

Financial Viability & Business Mode

Tiered Subscription Plans

Steady, predictable subscription revenues fuel foundational platform operations and feature enhancements.

Plan Level	Target Audience	Monthly Price
Explorer	Entry-level casual users	Free
Nomad	Regular adventure travelers	99 MAD
Elite	High-frequency explorers	199 MAD

Marketplace Revenue Model

Transactional commissions represent high-scale upsides tied directly to regional tourism development.

10% Commission Booking Fee: Added margin applied cleanly to all scheduled and processed guide events.

Dual Stream System: Generates highly stable predictable baseline revenues alongside variable high-volume peak-season transactional commissions.

Mutually Beneficial Ecosystem: Incentivizes high guide service quality while limiting platform financial exposure.

Cost Structure & Break-Even Analysis

Monthly Operating Costs

Operational costs are kept highly minimized to ensure extreme capital efficiency during our launch phase.

Component	Infrastructure Role	Estimated Cost
Hosting (Vercel)	Serverless application frontend	~ 200 MAD
Database (Supabase)	Relational content cloud storage	~ 250 MAD
Maps & Panoramas	Immersive mapping visual APIs	~ 500 MAD
AI Generation	Personalized smart itinerary model	~ 150 MAD

Path to Profitability

Minimal monthly subscription metrics or baseline booking transactions satisfy all infrastructural overheads.

11

NOMAD USERS
Active monthly memberships

8

COMPLETED BOOKINGS
Monthly marketplace guides

This highly optimal cost-to-break-even profile demonstrates that the technological infrastructure remains lightweight, extremely sustainable, and agile in its early deployment stages.

Development
Environment and Tools

Démonstration du Plateforme de
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Testing & Quality Assurance

Unit Testing (Vitest)

- Pricing logic accuracy
- Commission fee math
- Refund rule pathways
- Core business rule states

Integration Testing

- Booking workflows
- Payment integrations
- API endpoint gateways
- Database data integrity

End-to-End (Playwright)

- Subscription upgrade flow
- Guide service lifecycles
- Multi-tenant auth checks
- Responsive viewport tests

Additional Platform Engineering Quality Measures

Strict TypeScript Safety:
Static-typed interfaces prevent
variable bugs at runtime.

Modular Application: Decoupled
architectures allow fast, agile
features development.

Continuous Validation: Integrity
checks embedded directly into Git
deployment lifecycles.



Conclusion & Future Perspectives



Atlas 360 demonstrates the feasibility of a Morocco-first travel marketplace that unifies itinerary planning, immersive destination exploration, and secure guide booking. Beyond feature development, the project addressed real-world software engineering challenges such as transaction security, concurrency management, access control, and AI integration. The final result is a robust, scalable, and extensible platform with strong potential for future growth and commercialization.



Conclusion & Future Perspectives



Future Perspectives

- Integration of Moroccan payment gateways (CMI / PayZone)
- Native mobile applications (iOS & Android)
- Real-time messaging and push notifications
- B2B partnerships with Riads, agencies, and tourism operators
- Advanced recommendation engine based on traveller behaviour
- Expansion of guide verification and certification processes

**Thank You For Your
Attention**